

IT5106 - Software Development Project

Project Proposal - 2025

(Submission deadline: 24th November 2025)

Candidate Details

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Supervisor/Advisor Details

	Supervisor 1 (IT Related)	Supervisor 2 (Optional)
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Academic/ professional/ qualifications and memberships	PhD MIEEE	
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Note: Any deviation of the final project from the project described in this proposal should be explained by the candidate in the final Project Report.

Project Details

1. Title of Project:

Web-Based Vehicle Auction Platform

2. Name and address of client (ONLY if applicable):

Client	-	Tapro Japan Co. Ltd.
Address	-	1-1-14, Kamiikedai, Ota-Ku, Tokyo, Japan.
Postal Code	-	145-0064
Email	-	info@taprojapan.co.jp
Tel	-	+819096895141

3. Brief Introduction:

This project aims to design and develop a web-based vehicle auction and bidding platform for a vehicle exporter operating from Japan. The client purchases vehicles from weekly Japanese auto auctions and exports them to international buyers. Currently, the process of sharing vehicle details, collecting buyer interest, and identifying the highest bidder is performed manually through messaging platforms and unstructured communication channels. These manual methods are time-consuming, inefficient, and prone to misunderstandings regarding prices, bid validity, and the identity of the highest bidder.

The proposed system will centralize and automate the entire workflow by providing an online platform where users can browse newly arrived auction vehicles, search and filter listings, and place bids. Users must register an account to be eligible for bidding, and only verified accounts are allowed to bid. This ensures that only trusted and approved users can participate in auctions. The admin can add new vehicles, manage ongoing auctions, verify buyers, view bidding history, and select the winning bid based on the highest bid or other preferences determined by the business.

The system will be implemented as a full-stack web application using React for the front end, Node.js/Express for the backend API, and a SQL database for secure storage of vehicle information, users, and bidding data. The platform is designed to improve efficiency, accuracy, transparency, and scalability in managing cross-border vehicle sales.

4. Motivation for project:

The client currently handles vehicle listings and buyer communications manually using social media messages and phone calls. As the number of vehicles and international buyers grows, this process becomes increasingly difficult to track and is prone to errors such as missed bids, duplicate bid entries, miscommunication about vehicle information, and disputes regarding who placed the highest offer. These issues reduce efficiency and can negatively affect customer satisfaction.

Additionally, there is no structured mechanism to restrict bidding to verified and trusted buyers. As a result, unverified users may place bids without proper commitment, causing delays in decision-making and risking financial losses. The absence of a centralized system also makes it challenging for the client to maintain accurate historical records, identify serious buyers, or analyze bidding trends over time.

The client currently serves only a small number of customers and now intends to expand his business. To take his operation to the next level, he requires a professional, scalable, and automated digital platform that can attract more international buyers and manage increasing business volume without operational bottlenecks.

A dedicated web-based auction platform will significantly reduce manual work, streamline weekly auction vehicle updates, and ensure that only verified buyers can place valid bids. This will improve operational efficiency, enhance transparency, strengthen customer trust, and provide the client with a modern online presence that supports business growth. From an academic perspective, the project offers an excellent opportunity to model a real-world business process and implement it using modern web technologies.

5. Project Objectives

1. To develop a web-based platform that streamlines the client's vehicle auction process and improves transparency for buyers.
2. To ensure secure and trustworthy participation by implementing proper user verification and controlled bidding access.
3. To support efficient auction management by providing administrators with tools for handling vehicles, users, auctions, and winning bidder selection.
4. To maintain accurate historical records and ensure system reliability, data integrity, and secure handling of all interactions.
5. To refine the system through stakeholder feedback to better align with business needs.

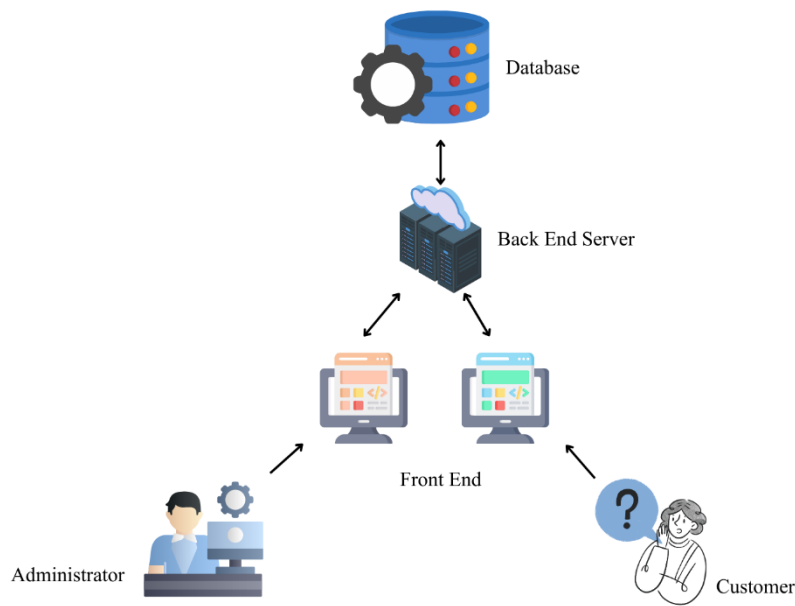


Figure 1 System Architecture of the Vehicle Auction Platform

This diagram depicts the three-tier architecture consisting of the database layer for data storage, the backend server handling business logic and API operations, and the frontend layer providing separate interfaces for administrators (vehicle and auction management) and customers (browsing and bidding functionality).

6. Scope of proposed project:

The project will cover the full development cycle of a web-based vehicle auction platform. The system will allow public users to browse weekly vehicle listings with search, filtering, and detailed vehicle pages. Registered users can place bids on active auctions but only after their accounts are verified by the admin.

Administrator will have separate privileged access to:

1. Add and update vehicle listings
2. Upload weekly auction vehicles
3. Manage auctions, start/end dates, and bid settings
4. Verify or reject buyer accounts
5. View bid history and select winning bidders
6. Monitor active bids and preferences

The system will include the following:

1. Secure login
2. Validation of bidding rules
3. Basic reporting functions

The project scope does not include payment integration, shipping logistics management, or advanced analytics; however, these may be considered as future enhancements.

7. Critical functionalities for project:

1. User registration and account verification workflow
2. Secure authentication and role-based access control
3. Vehicle listing management (CRUD)
4. Auction scheduling with start/end times
5. Bidding module for verified users
6. High-bid detection and bid history tracking
7. Admin dashboard for verification and decision-making
8. Search and filter functionality for buyers
9. Basic reporting (bid counts, winning bidders, auction results)

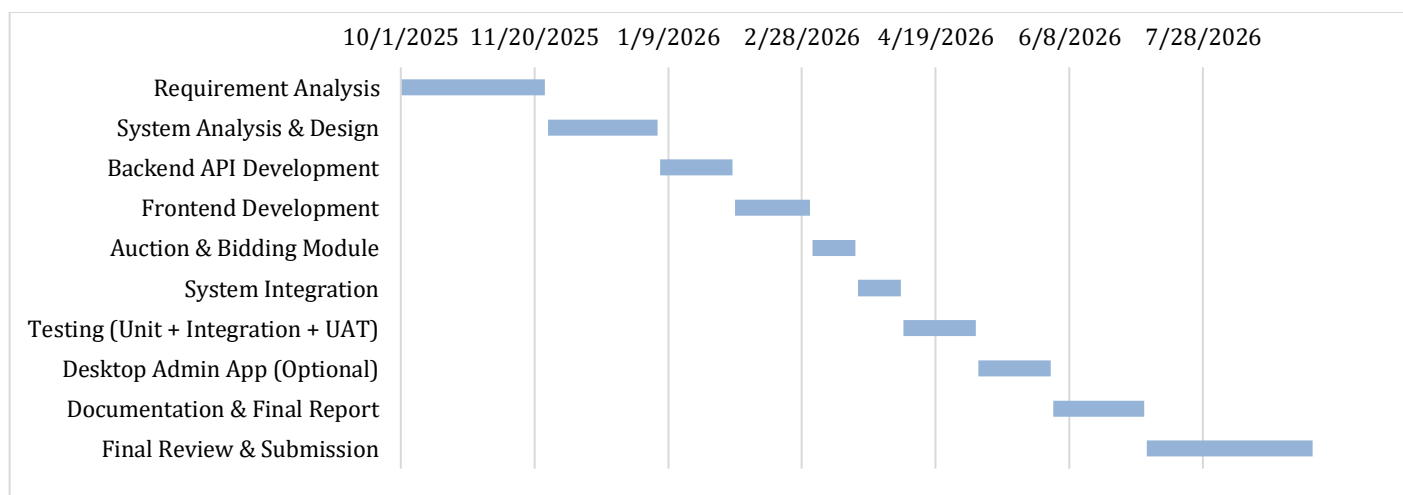
8. Itemized list of deliverables of the system:

Note: Deliverables are items that you would deliver to the client at the end of the project.

1. Software Requirements Specification (SRS)
2. System design documentation (use case diagrams, activity diagrams, class diagrams, etc.)
3. Fully implemented React frontend application
4. Fully implemented Node.js/Express backend with REST APIs
5. SQL database schema and implementation
6. Admin dashboard (web interface)
7. User interface with browsing, account creation, and bidding
8. Testing documentation (unit, integration, UAT)
9. Deployment guide or local installation manual
10. Final project report
11. User manual for the client

9. A project plan using Gantt chart (including the work involved in system development as well as writing the Final Project Report):

Table 1 Software Development Project Plan (Gantt Chart)



10. Resource requirements for project :

Functional Requirements:

1. User Management
 - a. User registration and login with secure authentication
 - b. Admin verification of user accounts before allowing bidding
 - c. Role-based access control (Admin vs. Customer)
2. Vehicle Management
 - a. Add, update, and delete vehicle listings by admin
 - b. Display vehicle details (images, specifications, auction date)
 - c. Search and filter vehicles by make, model, year, price range
3. Auction Management
 - a. Create auctions with start and end dates/times
 - b. Automatic auction status updates (active, closed, upcoming)
 - c. Weekly batch upload of new auction vehicles
4. Bidding System
 - a. Allow verified users to place bids on active auctions
 - b. Real-time bid validation (minimum bid increments, auction timing)
 - c. Track and display current highest bid
 - d. Prevent bidding after auction closure
5. Admin Dashboard
 - a. View all pending user verification requests
 - b. Monitor active bids and bidding history
 - c. Select winning bidder for each auction
 - d. Generate reports on auction results and user activities
6. Notifications and Alerts
 - a. Notify users of account verification status
 - b. Alert users when outbid on a vehicle
 - c. Inform winning bidders of auction results
7. Security & Data Management
 - a. Secure password storage with encryption
 - b. Session management and timeout handling
 - c. Data backup and recovery mechanisms

Technical Requirements:

1. React.js
2. Node.js and Express
3. PostgreSQL
4. VS Code
5. Git version control
6. Postman API testing tool
7. Diagramming software (Draw.io)
8. Web browser (Chrome/Edge)

11. Proposed way of self-evaluating the success of your system:

1. Compare functional requirements with implemented features using a completeness checklist.
2. Validate that only verified users can successfully place bids.
3. Test auction logic for correctness (start/end times, highest bid selection).
4. Test admin workflows (vehicle upload, user verification, bid assessment).
5. Conduct user acceptance testing with the client and selected buyers.
6. Evaluate the reliability and performance of the system under expected load.
7. Review system usability based on feedback from real users.
8. Confirm data consistency and database integrity through test cases.

12. Risk Assessment and Mitigation

This section identifies potential risks that may impact the successful delivery of the web-based vehicle auction platform and outlines corresponding mitigation strategies to minimize their likelihood and impact.

Risk	Impact	Mitigation Strategy
Real-time bidding validation challenges (concurrent bids, timing enforcement)	High	Use established libraries (Express-validator, Socket.io), implement database transactions, allocate 1-2 weeks buffer for testing
Data integrity and security issues for bidding data and user credentials	High	Use bcrypt encryption, parameterized queries, PostgreSQL constraints, regular backups, security audits
Timeline delays in complex auction module development	High	Break into sub-tasks, prioritize core features, allocate 15-20% contingency time, bi-weekly progress reviews
Technology learning curve (React, Express, PostgreSQL)	Medium	Complete tutorials early, leverage documentation, consult supervisor for key decisions
Scope creep due to additional client feature requests	Medium	Maintain clear scope of documentation, defer extras to Phase 2, require supervisor approval for changes